Supplementary Material for HARP-Select

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1 Purpose of the Supplement

This supplement expands the anonymous main paper with details that do not fit in the seven-page AAAI submission format. It is a frozen artifact companion: the tables below are generated from the included CSV files, and the verifier scripts check that the reported values match those CSVs.

The main claim remains deliberately bounded. HARP-Select is not presented as a universal heterophily leaderboard method. It is an auditable way to compare a self-loop residual polynomial expert with an ego-separated no-self expert and to record when validation evidence justifies switching between them.

2 Algorithmic Details

Residual polynomial experts. Both experts use the same residual polynomial filtering template, but they differ in how ego information enters propagation. The self-loop expert uses the normalized adjacency with self-loops. The ego-separated expert removes self-loops from propagation and carries ego features through a separate path before residual filtering. This makes the branch identity a testable modeling choice rather than an implicit preprocessing convention.

Selector rule. For each split, both experts are trained independently using the prescribed training and validation masks. Let v_{self} and v_{esep} be their validation scores, and let τ be the fixed normal-approximation uncertainty threshold used in the main paper. The selector chooses

$$\text{select} = \begin{cases} HARP-ESep, & v_{\text{esep}} - v_{\text{self}} > \tau, \\ HARP-GNN, & \text{otherwise.} \end{cases}$$

The selected branch is frozen before test labels are read. The artifact logs the validation margin, threshold, selected branch, oracle branch, and regret.

Review-time audit procedure. The intended audit sequence is:

1. Verify that every reported config/result grid is complete.
2. Rebuild all generated tables from CSV files.
3. Check that paper text and table values match the rebuilt results.
4. Check that statistical claims do not exceed paired-test evidence.
5. Inspect selector decisions and oracle regret split by split.

3 Filter-Response Interpretation

The residual branch has a direct polynomial-filter interpretation. Suppose the normalized self-loop operator has eigendecomposition $\hat{A} = Q\Lambda Q^{\top}$. For any feature column, the low-pass branch applies

$$p_L(\lambda) = \sum_{k=0}^{K} \alpha_k \lambda^k,$$

where the softmax coefficients keep the response as a convex combination of adjacency powers. The residual branch applies

$$p_H(\lambda) = \sum_{k=1}^{K} \beta_k(\lambda^{k-1} - \lambda^k) = (1-\lambda)\sum_{k=1}^{K} \beta_k \lambda^{k-1}.$$

Thus the residual response is exactly zero on the constant eigencomponent $\lambda = 1$ and increases on components that are attenuated by repeated smoothing. This is the sense in which the HARP branch is high-pass: it is not an unconstrained extra MLP, but a finite-difference complement to a smoothing polynomial.

This interpretation also clarifies why branch identity matters. The self-loop expert measures residuals relative to a propagation operator that always carries ego information. The ego-separated expert instead propagates with a no-self operator and concatenates the original ego state only at the readout. Consequently, the two branches are not merely different parameterizations of the same filter. They encode different assumptions about whether self information should be mixed into every propagation step or kept as a separate reference signal.

Selector calibration caveat. The main-paper threshold uses a normal-approximation standard error for two validation accuracies. The validation predictions are paired on the same nodes, so the independence approximation is not exact. We intentionally use this simple threshold as a conservative branch-switching rule rather than as a formal hypothesis test. All paired significance claims in the paper are made on held-out test split results, not on the validation routing statistic.

4 Extended Main Tables

4.1 HARP-Select Summary

Table 1: HARP-Select summary over fixed splits.

Dataset	HARP-GNN	HARP-ESep	HARP-Select	Oracle	ESep splits
actor	35.08 ± 1.36	35.62 ± 1.09	35.08 ± 1.36	35.86 ± 0.92	0/10
amazon-ratings	45.43 ± 0.71	45.94 ± 0.48	45.43 ± 0.71	46.18 ± 0.31	0/10
chameleon	55.90 ± 2.49	63.42 ± 2.24	63.42 ± 2.24	63.42 ± 2.24	10/10
cornell	74.86 ± 4.60	68.92 ± 5.87	74.86 ± 4.60	75.68 ± 4.41	0/10
roman-empire	75.76 ± 0.63	78.51 ± 0.57	78.51 ± 0.57	78.51 ± 0.57	10/10
squirrel	36.66 ± 1.66	46.48 ± 1.10	46.48 ± 1.10	46.48 ± 1.10	10/10
texas	86.49 ± 4.59	75.41 ± 4.84	86.49 ± 4.59	86.49 ± 4.59	0/10
wisconsin	87.45 ± 3.23	82.16 ± 4.57	87.45 ± 3.23	87.45 ± 3.23	0/10

Table 2: HARP-Select versus the strongest implemented non-HARP baseline on the original six fixed-split heterophily datasets.

Dataset	Best baseline	Baseline	HARP-Select	Diff (pp)	p
actor	LINKX	35.67	35.08	-0.59	0.245
chameleon	H2GCN	64.21	63.42	-0.79	0.395
cornell	MLP	76.49	74.86	-1.62	0.405
squirrel	LINKX	45.48	46.48	+1.01	0.273
texas	MixHop	84.59	86.49	+1.89	0.132
wisconsin	H2GCN	85.49	87.45	+1.96	0.085

Table 3: Robust split-level checks for HARP-Select on the original six fixed-split heterophily datasets.

Dataset	Baseline	Diff (pp)	Bootstrap 95% CI	W/T/L	Sign-flip p	Holm p
actor	LINKX	-0.59	[-1.49, +0.24]	3/1/6	0.242	1.000
chameleon	H2GCN	-0.79	[-2.43, +0.86]	4/0/6	0.383	1.000
cornell	MLP	-1.62	[-5.14, +1.62]	3/3/4	0.359	1.000
squirrel	LINKX	+1.01	[-0.53, +2.72]	6/0/4	0.299	1.000
texas	MixHop	+1.89	[-0.27, +4.05]	5/3/2	0.203	1.000
wisconsin	H2GCN	+1.96	[+0.00, +3.73]	7/1/2	0.125	0.750

4.2 External Critical-Heterophily Accuracy Benchmarks

Table 4: Full external critical-heterophily comparison including implemented non-HARP baselines and HARP branches.

Dataset	MLP	GCN	SGC	APPNP	MixHop	GPR-GNN	H2GCN	LINKX	HARP-GNN	HARP-ESep	HARP-Select
amazon-ratings	38.79 ± 0.30	42.76 ± 0.42	38.77 ± 0.49	42.28 ± 0.39	43.88 ± 0.54	44.32 ± 0.31	44.68 ± 0.45	**46.34 ± 0.81**	45.43 ± 0.71	45.94 ± 0.48	45.43 ± 0.71
roman-empire	64.46 ± 0.54	41.58 ± 0.62	35.84 ± 0.50	49.45 ± 0.49	77.22 ± 0.50	69.45 ± 0.34	77.73 ± 0.68	60.42 ± 0.90	75.76 ± 0.63	**78.51 ± 0.57**	**78.51 ± 0.57**

Table 5: Paired checks against the strongest implemented external non-HARP baseline.

Dataset	Target	Baseline	Diff (pp)	Bootstrap 95% CI	W/T/L	Holm p
amazon-ratings	HARP-ESep	LINKX	-0.41	[-1.03, +0.18]	3/0/7	0.254
amazon-ratings	HARP-Select	LINKX	-0.91	[-1.34, -0.40]	1/0/9	0.020
roman-empire	HARP-ESep	H2GCN	+0.79	[+0.61, +0.99]	10/0/0	0.008
roman-empire	HARP-Select	H2GCN	+0.79	[+0.61, +0.98]	10/0/0	0.008

4.3 Binary Critical-Heterophily ROC-AUC

Table 6: Complete binary critical-heterophily branch comparisons under ROC-AUC.

Dataset	HARP-GNN	HARP-ESep	Diff (pp)	n	p
minesweeper	88.18 ± 0.84	89.64 ± 0.67	+1.45	10	< 0.001
tolokers	82.79 ± 0.83	79.22 ± 0.70	-3.56	10	< 0.001

Table 7: Robust binary critical-heterophily branch checks under ROC-AUC.

Dataset	Diff (pp)	Bootstrap 95% CI	W/T/L	Sign-flip Holm p
minesweeper	+1.45	[+1.27, +1.67]	10/0/0	0.004
tolokers	-3.56	[-3.96, -3.19]	0/0/10	0.004

5 Selector Diagnostics

Table 8: Threshold sensitivity for the validation-calibrated selector.

z	ESep splits	Macro acc.	Regret (pp)	Oracle match
0	48/80	64.59	+0.42	88.8
0.5	41/80	64.87	+0.14	91.2
1	34/80	64.75	+0.26	83.8
1.645	31/80	64.71	+0.30	80.0
1.96	30/80	64.72	+0.29	81.2
2.58	26/80	64.36	+0.65	76.2

Table 9: CPU training cost for two-expert routing.

Dataset	HARP sec.	ESep sec.	Two-expert sec.	Cost / HARP
actor	26.0	15.1	41.1	1.58×
amazon-ratings	182.7	116.2	298.9	1.66×
chameleon	20.7	28.1	48.8	2.38×
cornell	4.4	0.9	5.3	1.24×
roman-empire	83.7	90.0	173.7	2.11×
squirrel	86.8	92.5	179.2	2.18×
texas	4.4	1.3	5.7	1.34×
wisconsin	5.6	1.4	7.0	1.25×

Interpretation. The selector is intentionally conservative. This avoids converting small validation fluctuations into test-time architecture search, but it can miss small real branch advantages. Amazon-Ratings is the clearest example: the ego-separated branch is slightly better on average, yet validation evidence is not strong enough to trigger switching. Tolokers is the opposite binary stress test: removing ego/self information is consistently harmful.

6 Mechanistic Ablations

Table 10: WebKB ablations isolating multi-hop mixing, low/high-pass residual branches, and the gate signal.

Dataset	MixHop	HARP-GNN	HARP-Low	HARP-High	HARP-NoSignal
cornell	74.59 ± 3.17	74.86 ± 4.60	51.35 ± 11.18	72.43 ± 6.34	**75.41 ± 4.12**
texas	84.59 ± 4.60	**86.49 ± 4.59**	62.43 ± 7.69	81.35 ± 6.43	85.14 ± 6.40
wisconsin	83.73 ± 5.47	**87.45 ± 3.23**	75.49 ± 6.22	79.02 ± 4.90	**87.45 ± 4.26**

Table 11: Learned HARP polynomial weights on WebKB.

Dataset	L0	L1	L2	L3	H1	H2	H3
cornell	0.317 ± 0.015	0.236 ± 0.017	0.223 ± 0.010	0.223 ± 0.010	0.473 ± 0.054	0.265 ± 0.025	0.263 ± 0.029
texas	0.301 ± 0.017	0.263 ± 0.023	0.218 ± 0.012	0.218 ± 0.010	0.481 ± 0.047	0.262 ± 0.022	0.257 ± 0.025
wisconsin	0.316 ± 0.012	0.227 ± 0.009	0.230 ± 0.008	0.227 ± 0.005	0.459 ± 0.022	0.277 ± 0.016	0.264 ± 0.008

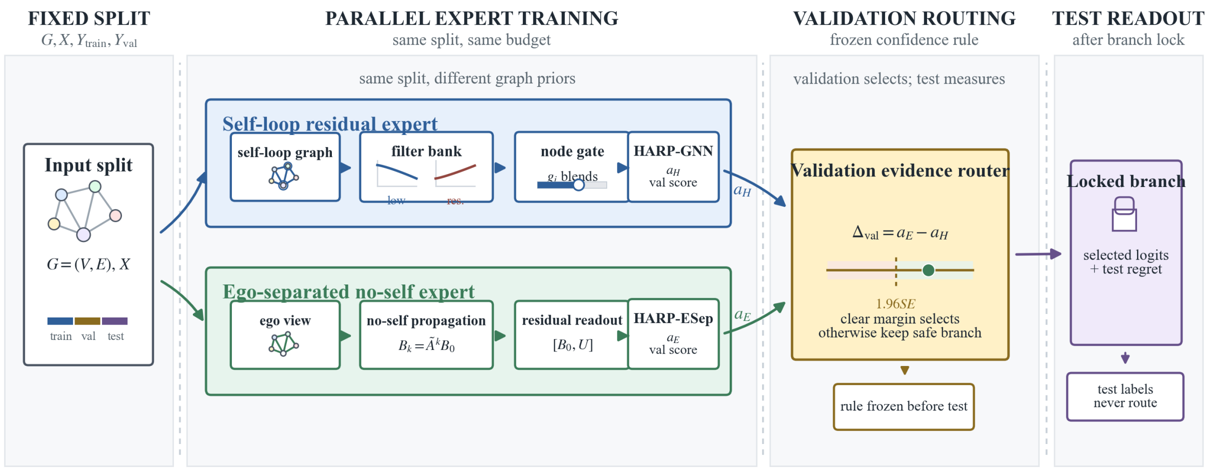

Figure 1: HARP-Select trains two experts, records validation evidence, and freezes the branch decision before test evaluation.

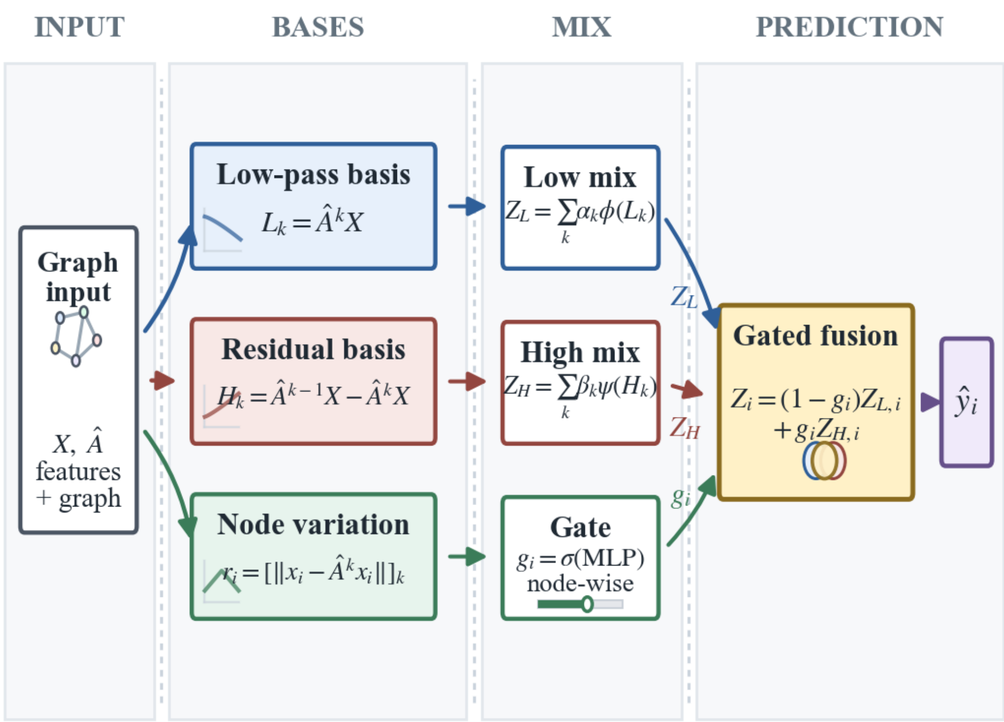

Figure 2: Internal residual polynomial block used by the self-loop expert.

7 Reproducibility Checklist for Reported Numbers

The following scripts are the minimal checks for the numbers reported in the main paper and this supplement:

```
python scripts\verify_implementation.py
python scripts\verify_manuscript_integrity.py
python scripts\verify_reported_results.py
python scripts\verify_top_conference_claims.py
python scripts\verify_anonymous_artifact.py --root .
```

The full packaging check additionally validates the official PDF, checklist, source package, supplementary artifact, synchronized packed sources, and LaTeX logs:

```
python scripts\verify_submission_readiness.py
```

8 Result Coverage Audit

Table 12: Config/result coverage audit. Each row must be complete before a result is treated as reported evidence.

Config	Status	Expected	Observed	Missing	Extra	Dup.
configs\ablation_synthetic.yaml	complete	24	24	0	0	0
configs\actor_smoke.yaml	complete	3	3	0	0	0
configs\cora.yaml	complete	15	15	0	0	0
configs\critical_heterophily_baselines.yaml	complete	160	160	0	0	0
configs\critical_heterophily_binary_harp.yaml	complete	40	40	0	0	0
configs\critical_heterophily_binary_smoke.yaml	complete	6	6	0	0	0
configs\critical_heterophily_fagcn_style_smoke.yaml	complete	2	2	0	0	0
configs\critical_heterophily_harp.yaml	complete	40	40	0	0	0
configs\critical_heterophily_smoke.yaml	complete	4	4	0	0	0
configs\geom_gcn_harp_esep.yaml	complete	30	30	0	0	0
configs\geom_gcn_large.yaml	complete	270	270	0	0	0
configs\geom_gcn_smoke.yaml	complete	12	12	0	0	0
configs\harp_x_diagnostic.yaml	complete	96	96	0	0	0
configs\planetoid_all.yaml	complete	75	75	0	0	0
configs\planetoid_projected_smoke.yaml	complete	3	3	0	0	0
configs\planetoid_smoke.yaml	complete	9	9	0	0	0
configs\smoke.yaml	complete	3	3	0	0	0
configs\synthetic_sweep.yaml	complete	45	45	0	0	0
configs\webkb.yaml	complete	270	270	0	0	0
configs\webkb_ablation.yaml	complete	150	150	0	0	0
configs\webkb_fagcn_style_smoke.yaml	complete	3	3	0	0	0
configs\webkb_gate_variants.yaml	complete	120	120	0	0	0
configs\webkb_h2gcn.yaml	complete	30	30	0	0	0
configs\webkb_h2gcn_smoke.yaml	complete	1	1	0	0	0
configs\webkb_harp_diagnostics.yaml	complete	30	30	0	0	0
configs\webkb_harp_esep.yaml	complete	30	30	0	0	0
configs\webkb_scalar_gate.yaml	complete	150	150	0	0	0
configs\webkb_smoke.yaml	complete	6	6	0	0	0